

IBM University Engagement Project Submission

Street Vendor Digitalization Agent

Project Tile - **Street Vendor Digitalization Agent**

Domain of Project - **Financial Inclusion**

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Problem Statement

► Street Vendor Digitalization Agent

► The Challenge

- Millions of street vendors and micro-entrepreneurs across India operate entirely offline, and struggle with **business visibility, digital payments, and access to formal credit**. Information about relevant government schemes, listing platforms, and payment setup is scattered across many portals and rarely reaches vendors directly. Without **personalized, real-time guidance**, many vendors remain excluded from the digital and formal economy.

► The Objective

- Build an **AI-powered Street Vendor Digitalization Agent** that delivers **one-sentence-to-storefront digital onboarding** for street vendors and micro-entrepreneurs. The solution should:
 - **Personalized Business Profile** - Generate a business name, bio, and listing text from one plain-language sentence, in the vendor's own language.
 - **Local Scheme & Platform RAG Retrieval** - Fetch and summarize live details from PM SVANidhi, ONDC, and Google Business Profile onboarding sources.

- **Digital Payment Enablement** - Guide vendors step-by-step through UPI QR code setup so they can accept digital payments immediately.
- **Hyperlocal Promotion** - Suggest pricing tips, local SEO, and promotional content suited to the vendor's exact trade and area.
- **Multi-Agent System** -
 - **Business Profile Agent** (generates name, bio, and listing text)
 - **Scheme & Listing Agent** (retrieves scheme and platform data via RAG)
 - **Payment Enablement Agent** (walks through UPI QR setup)
 - **Promotion Agent** (pricing tips and local promotion ideas)
- **Vendor Dashboard** - Show the generated profile, saved action plan, and a quick-start checklist for going digital.

Proposed Solution

- ▶ **Proposed Solution: AI-Powered Street Vendor Digitalization Agent**
- ▶ The proposed solution is a **multi-agent AI-powered Street Vendor Digitalization Agent** built using **IBM watsonx Orchestrate**, **IBM watsonx.ai**, and **Granite Models** to take a vendor from a single spoken sentence to a usable digital storefront. The system addresses gaps in business visibility, scheme access, and digital payment adoption through an intelligent, modular architecture.
- ▶ At its core, the solution uses a **Retrieval-Augmented Generation (RAG)** pipeline to fetch accurate, current information from trusted sources such as PM SVANidhi, ONDC, and Google Business Profile. This data is embedded and stored in a vector database, enabling the system to retrieve and ground scheme, listing, and payment guidance using Granite models.
- ▶ The system consists of four key agents:
- ▶ **Business Profile Agent** generates a business name, bio, and listing text from the vendor's plain-language input.
- ▶ **Scheme & Listing Agent** retrieves relevant government schemes and platform onboarding steps using RAG.

- ▶ **Payment Enablement Agent** walks the vendor through UPI QR code setup for digital payments.
- ▶ **Promotion Agent** generates pricing tips and hyperlocal promotion ideas suited to the vendor's trade and area.
- ▶ The solution includes a **simple, card-based vendor dashboard** (built with React) that displays the generated profile, action plan, and a quick-start checklist for going digital.
- ▶ IBM watsonx Orchestrate is used to orchestrate agent workflows, while watsonx.ai ensures scalable model deployment, embeddings, and AI governance. Granite models power reasoning, summarization, and personalization.
- ▶ Overall, this solution delivers an **end-to-end intelligent digitalization assistant** that helps street vendors gain visibility, access credit, and accept digital payments.

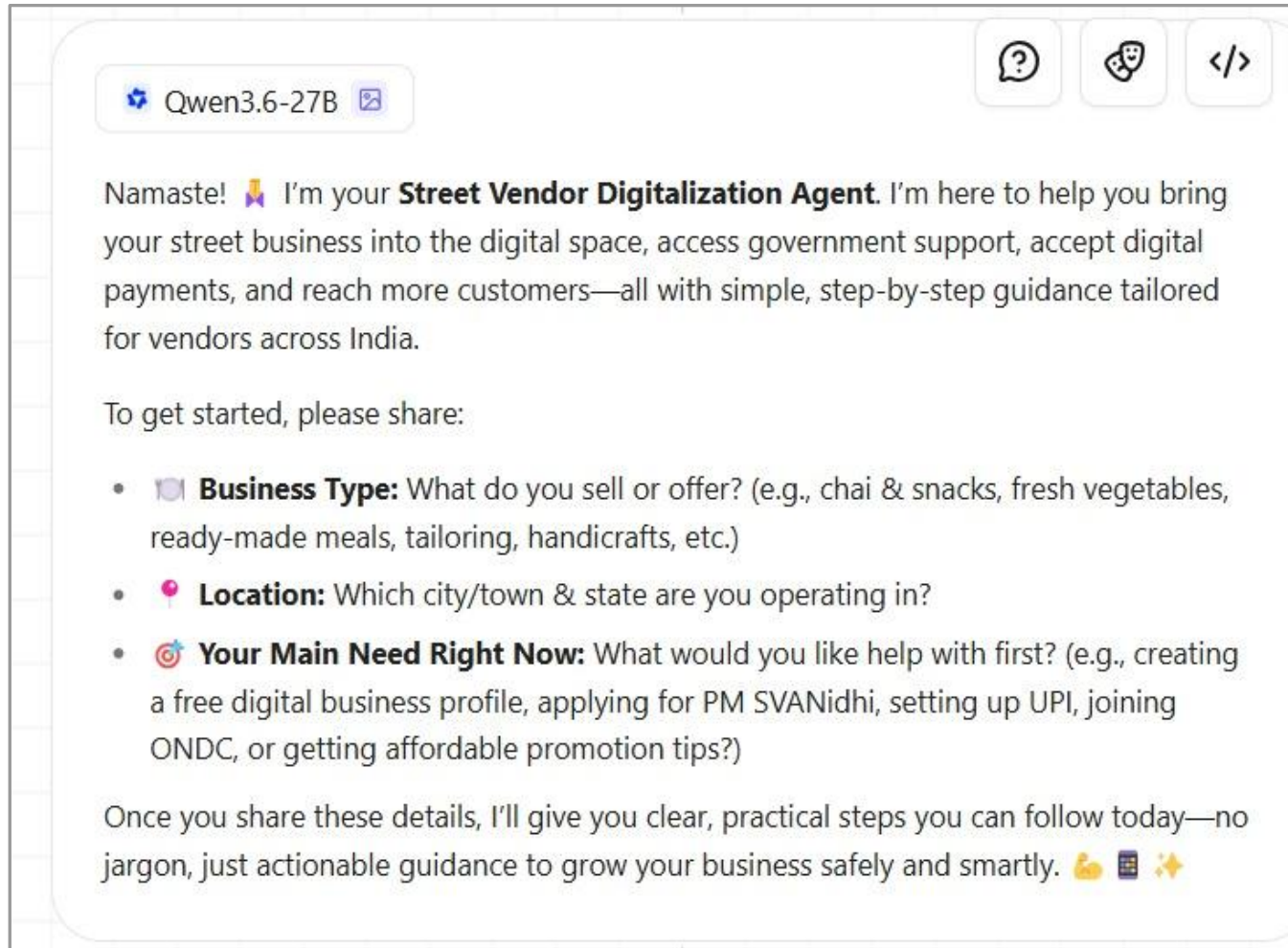
Technologies Used

- **IBM Granite model** - for eg - ibm-granite-3-2-8b - For natural language understanding, reasoning, and generating vendor profiles and guidance
- **IBM Cloud** - It provides the infrastructure to build, deploy, and scale the Street Vendor Digitalization Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the agent to thousands of vendors.
- **RAG**- RAG to fetch real scheme and platform data first, then generate grounded guidance, making the agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database (FAISS/Chroma)** - For RAG-based retrieval of scheme and platform documents

Orchestrate components used

- ▶ **1) Chat Input** - Interface where vendors enter their business description in plain language (text/voice).
- ▶ **2) IBM watsonx AI Agent** - Processes vendor input using Granite models to generate a grounded business profile and action plan.
- ▶ **3) Name of IBM Granite model** - Granite-3-2-8B-Instruct → Fast, efficient for real-time profile and guidance generation
- ▶ **4) Chat output** -Displays the generated business profile, pricing tips, listing steps, QR setup, and promotion ideas.
- ▶ **5) File component** - Stores and retrieves scheme documents, onboarding guides, and vendor session data for retrieval.
- ▶ **6) Web Search / Knowledge Base tool** - connects to PM SVANidhi, ONDC, and Google Business Profile sources for grounded retrieval.

Agent Instructions Screenshot



Document Uploaded Screenshot

Reference Sources Consulted

(No document knowledge base was uploaded to the agent — these public scheme resources informed the agent's built-in domain knowledge and instructions)

PM SVANidhi

PM Street Vendor's AtmaNirbhar Nidhi

Micro-credit scheme offering collateral-free working capital loans to street vendors.

pmsvanidhi.mohua.gov.in

ONDC

Open Network for Digital Commerce

Open e-commerce protocol enabling small vendors to list and sell digitally.

ondc.org

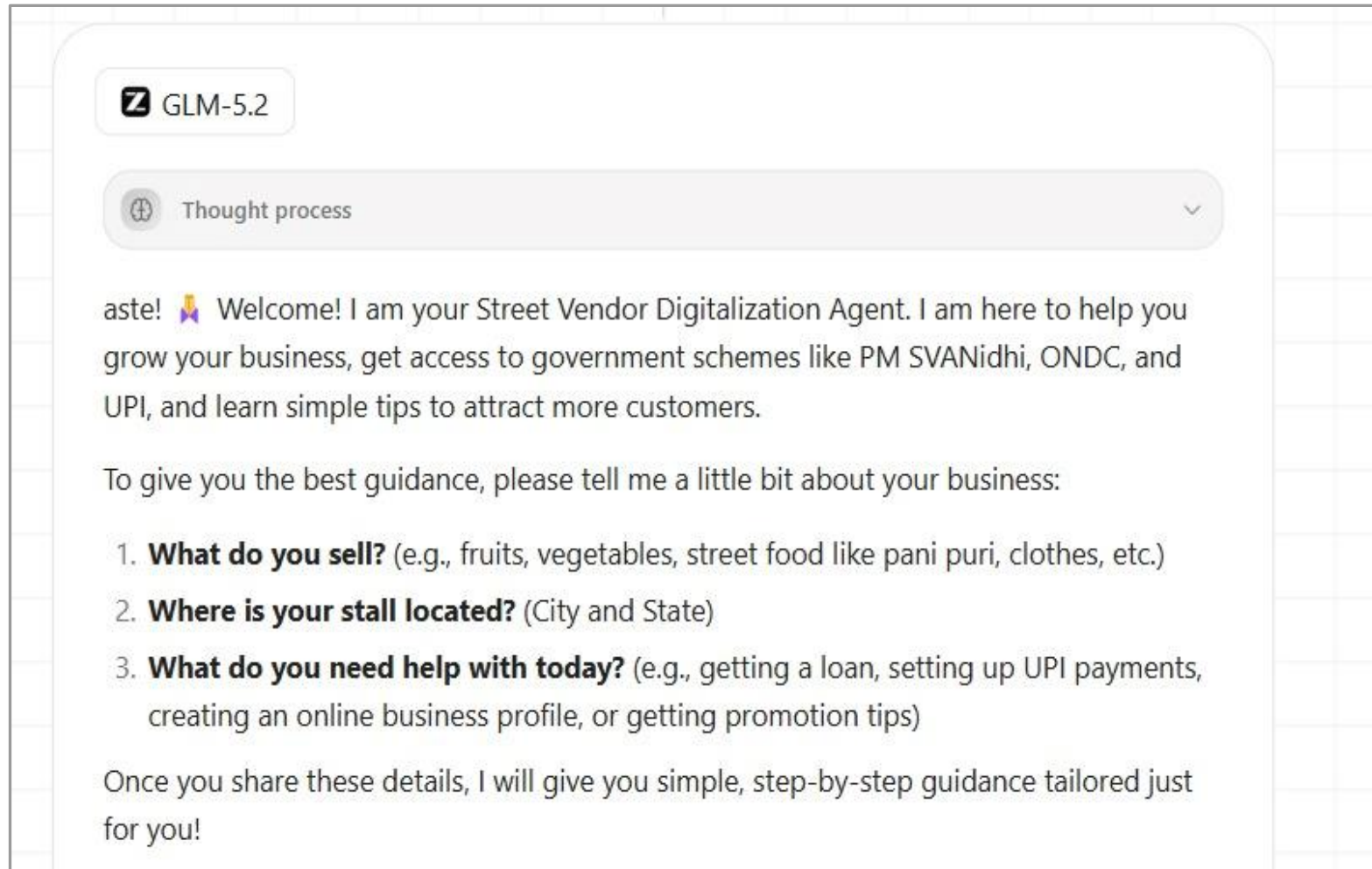
UPI

Unified Payments Interface

Real-time payment system enabling vendors to accept instant digital payments.

npci.org.in / bhimupi.org.in

Prompt with output Screenshot



Other Applicable Screenshots

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 Thought process



Hello! I'd be happy to help you understand how **PM SVANidhi** works for your chai & snack stall. First, a quick but important clarification: **PM SVANidhi is not a traditional bank loan you apply for directly.** Instead, it's an interest-free credit facility that the government automatically allocates to eligible informal self-employed workers once their profiles are verified and linked to a Jan Dhan Bank (JDB) account.

Here's a clear, step-by-step guide on how to check your eligibility, activate your credit, and track it specifically for Andheri, Mumbai:

 **Step 1: Check Your Basic Eligibility**

You may already be eligible if you meet these criteria:

- Engaged in informal self-employed activities (like running a tea/snack stall)
- Not under any formal employment (no regular contract, ESI/PF, or formal salary)
- Have a **Jan Dhan Bank (JDB) account** (linked to Aadhaar)
- Have an active **PM SVANidhi Card** (usually issued with a JDB card)
- Covered under the **mandatory insurance** provided by the scheme (₹10,000 life + ₹25,000 accidental)
- Not receiving wage/guide payments from MGNREGA or similar government wage programs

 **Step 2: Activate Your Credit in Andheri (Maharashtra)**

Since credit allocation is managed at the state/district level, here's what to do:

- 1. Verify Your Card & Status**
Visit the official portal: <https://svp.nic.in>
Enter your **Aadhaar number** or **PAN** to check if your SVANidhi card is active and linked.
- 2. Contact Your Ward Office or JDB Branch**
Andheri falls under the Mumbai Municipal Corporation. Visit your local **Ward Office** or your nearest **Jan Dhan Bank branch** (SBI, DBS, PNB, etc.).
Ask for the **"PM SVANidhi Credit Linkage Desk"** or **"Swajeeban Helpdesk"**. They can verify your stall's informal status and trigger credit allocation.



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- 3. Ensure Insurance & JDB Linking**
 - The scheme mandates insurance. If it's inactive, request your JDB branch to reactivate it.
 - Credit is only released after insurance, JDB, and SVANidhi card are successfully linked.
- 4. Wait for Credit Allocation**
 - Once verified, the government releases **₹25,000** (can be increased to ₹100,000 in later phases) directly into your JDB account. No interest is charged initially.

 **Important Things to Know**

- **No traditional application form:** Credit is automatically allocated once eligibility is confirmed through state surveys & bank linkage.
- **Interest & Repayment:** Initially interest-free. As per recent government notifications, after **5 years** it becomes repayable at **7% p.a. compounded annually** (subject to extension/renewal).
- **Track Your Status:** Use <https://svp.nic.in> or the **Swajeeban Maharashtra Portal** (<https://www.swajeeban.gov.in>) to check your status.
- **Documents to Keep Ready:** Shop license, GST registration (if any), utility bills, JDB passbook, Aadhaar card. These help during verification.

 **Official Contact (Maharashtra)**

- **Swajeeban Helpdesk:** 1914
- **Department of Social Justice & Employment (DSJE) Maharashtra:** Visit your district administration centre or ward office.
- **JDB Branch:** Ask specifically for the **"PM SVANidhi / Swamita Vaas Nidhi"** helpdesk counter.

 **Pro Tip for Stall Vendors**

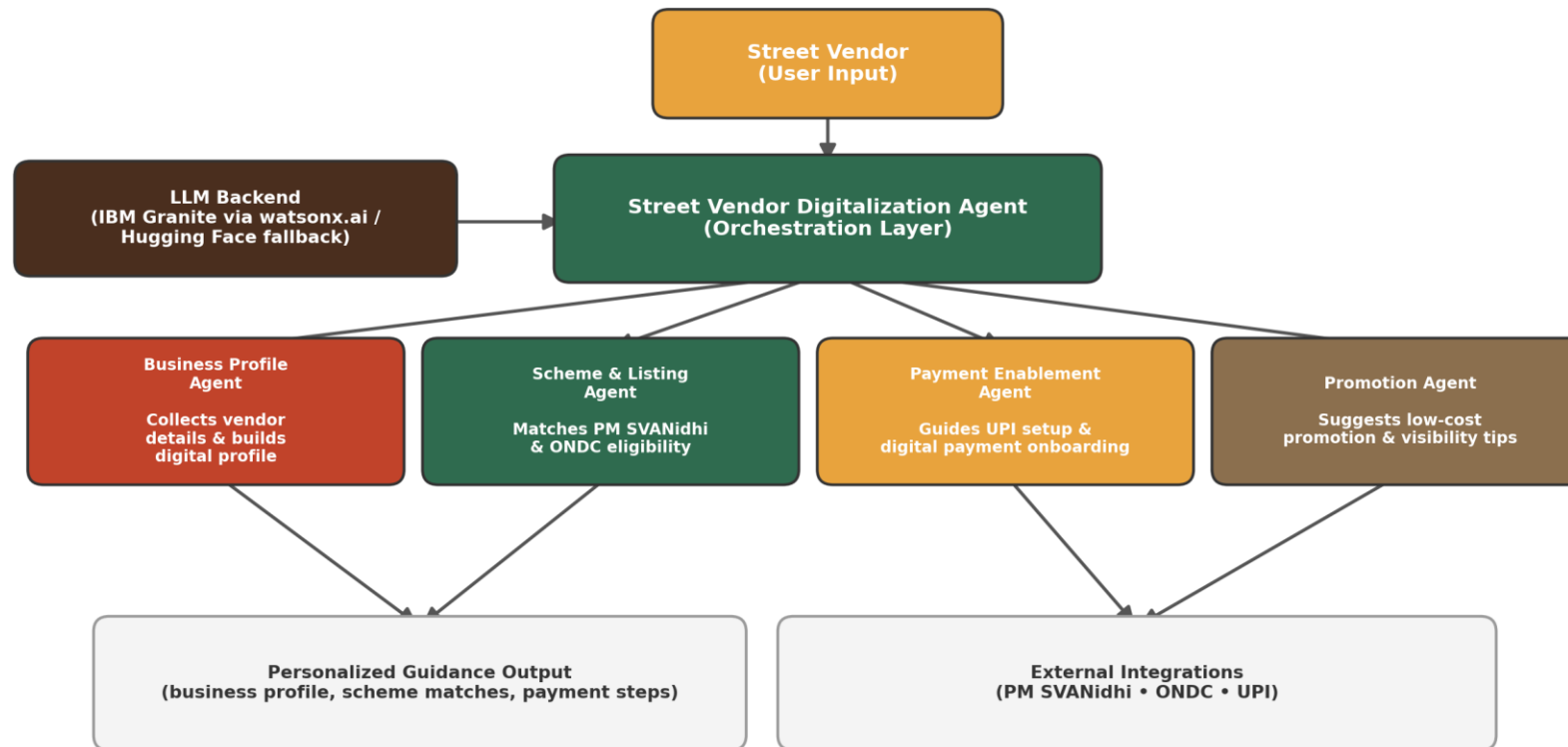
Many chai/snack stall owners in Mumbai were included during district-level Swamitva Vaas Nidhi surveys. If your credit hasn't shown up yet, it's likely pending verification or activation. A simple visit to your ward office or JDB branch with your JDB Aadhaar will clear this in 1–2 days.



Would you like help finding the exact ward office address, your nearest JDB SVANidhi desk, or a direct link to check your status with your Aadhaar/PAN? Just share the details

Orchestrate architecture blueprint

Street Vendor Digitalization Agent — Architecture



Role of Agentic AI in the solution

Role of Agentic AI in the Street Vendor Digitalization Agent

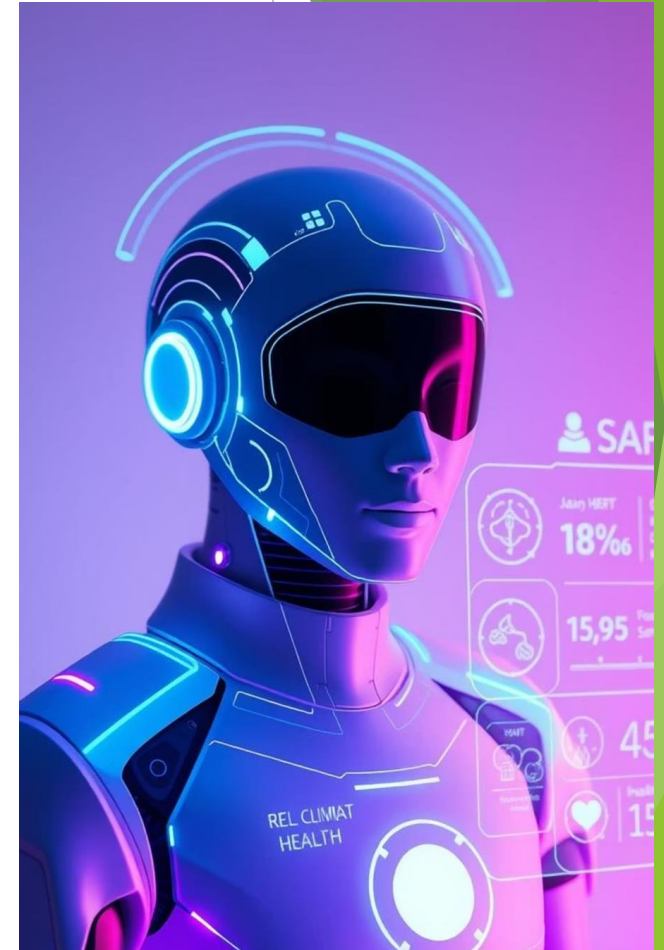
Agentic AI enables the Street Vendor Digitalization Agent to function as an intelligent, autonomous system rather than a simple chatbot. Using **IBM watsonx.ai** and **IBM Granite models**, it coordinates multiple specialized agents to take a vendor from one sentence to a usable digital storefront.

Each agent performs a specific role—such as generating a business profile, retrieving relevant schemes, enabling digital payments, and suggesting promotion—while working collaboratively. This **multi-agent approach** ensures efficient task execution and accurate, grounded results.

Agentic AI also brings **context awareness**, understanding the vendor's trade, location, and language to generate tailored guidance. It supports **real-time adaptation**, grounding every answer in the latest scheme and platform information.

Additionally, agents communicate with each other, enabling better decision-making and a coherent end-to-end action plan.

Overall, Agentic AI makes the system proactive, personalized, and goal-driven, acting like a **smart digital onboarding assistant** that helps vendors go digital in minutes, not weeks.



Novelty and Uniqueness

- ▶ The proposed Street Vendor Digitalization Agent stands out by combining **Agentic AI, RAG, and hyperlocal personalization** into a single intelligent system.
- ▶ **Multi-Agent Intelligence** - Instead of a single chatbot, four specialized agents collaborate to deliver a complete, actionable digital onboarding plan.
- ▶ **RAG-Based Accuracy** - Integrates real-time scheme and platform data retrieval with AI generation, reducing misinformation and ensuring trustworthy guidance.
- ▶ **Hyper-Personalization** - Tailors business names, pricing tips, and promotion ideas to the vendor's exact trade and location.
- ▶ **One-Sentence Onboarding** - Converts a single plain-language sentence into a complete digital storefront plan, removing the biggest usability barrier for low-digital-literacy users.

- ▶ **Multi-Language Ready** - Designed to accept and respond in regional Indian languages for real accessibility.
- ▶ **Financial Inclusion Focus** - Goes beyond visibility to connect vendors with credit-access schemes like PM SVANidhi.
- ▶ **Scalable & Enterprise-Ready** - Built on **IBM watsonx.ai** with **IBM Granite models**, ensuring reliability and governance for a national rollout.
- ▶ This makes the solution a **smart, adaptive digital onboarding partner**, not just a static profile generator.

Future Scope

1. Voice-First & WhatsApp Deployment –

The agent can be deployed as a WhatsApp bot with voice input support, reaching vendors who are not comfortable typing and giving near-universal access without requiring an app install.

2.Transaction-Based Credit Scoring –

By tracking UPI transaction history after onboarding, the agent can build a credit-readiness score, creating a direct bridge from digital payment adoption to formal micro-credit eligibility.

Inventory & micro-loan agents – extending the same Granite + RAG pattern to stock tracking and credit eligibility checks

Direct government portal integration – automated submission to PM SVANidhi and ONDC, removing manual steps.

Credly Certificate



IBM BOB Certificate

